ARB Basket Fair Value Research Engine v2

Whitepaper + Operating Tutorial

Internal research note and implementation guide for the upgraded ARB Basket Fair Value indicator. The objective is to reduce false ARB+ / ARB- prints by separating statistical dislocation from tradeable confirmation.

Version: Research Engine v2. Built from the current ARB Basket Fair Value structure engine codebase and upgraded with beta-adjusted fair value, beta stability, liquidity filters, session and bar-close controls, higher-timeframe trend, ADX/DMI trend-stress, and an expanded confluence score.

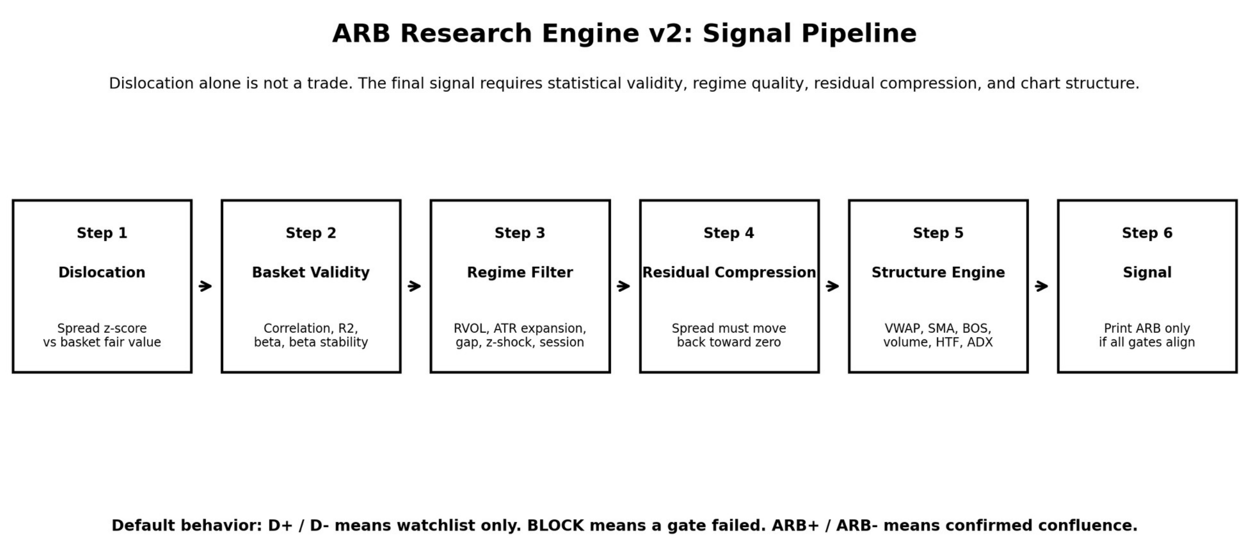

Figure 1. Research Engine v2 signal pipeline.

1. Executive Summary

The original ARB logic correctly identified relative-value dislocation, but several backtests showed that dislocation alone can create false positives. The IBM example was a typical failure mode: ARB+ printed while price was near or below the 200 SMA and failed at that structure instead of reclaiming it. The problem was not the idea of ARB. The problem was allowing a relative-value condition to behave like a complete trade signal.

Research Engine v2 changes the interpretation: D+ and D- are watchlist states, BLOCK is a diagnostic warning, and ARB+ / ARB- are reserved for confirmed setups only. A confirmed setup now requires statistical validity, regime quality, residual compression, VWAP/SMA/structure confluence, volume quality, optional higher-timeframe trend alignment, and optional ADX/DMI trend-stress protection.

Before	After
ARB+ could print from dislocation + limited confirmation.	ARB+ requires dislocation + valid basket + clear regime + compression + confluence score.
Fair value used raw basket return only.	Fair value can scale expected return by rolling beta.
Weak relationship could still influence signals.	Correlation, R-squared, beta, and beta stability are checked.
VWAP/SMA were secondary chart references.	VWAP/SMA can be critical gates before ARB prints.
Live-bar labels could flicker on realtime bars.	Optional bar-close confirmation reduces repaint-like behavior.

2. Research Foundation

2.1 Relative-value trading

The ARB model is closest to a multi-asset relative-value/pairs-trading framework. In the classic pairs-trading literature, a trade is entered when close substitutes diverge and exited when the spread converges. Gatev, Goetzmann, and Rouwenhorst test a relative-value arbitrage rule on daily U.S. equities and emphasize normalized price distance, convergence, transaction costs, and the role of close substitutes [1].

For this indicator, the basket replaces a single pair. The spread is the difference between the chart symbol return and the basket-implied expected return. This means ARB+ is not automatically bullish. ARB+ means the chart symbol is below basket-implied fair value. ARB- means it is above basket-implied fair value.

2.2 Stationarity and residual behavior

A statistical-arbitrage signal is stronger when the spread has some evidence of stationarity or mean reversion. Engle and Granger formalize cointegration as a condition where non-stationary time series can have stationary linear combinations [2]. A Pine indicator cannot run a full cointegration test cleanly with this architecture, so the implementation uses practical proxies: correlation, R-squared, beta, beta stability, z-score, and residual compression.

The key execution rule is: dislocation first, compression second, signal third. A high positive z-score only says the stock is cheap versus the basket. The long side becomes more credible only when the residual begins moving back toward zero.

2.3 VWAP, volatility, and trading conditions

VWAP is widely used as an intraday benchmark in execution practice. CFA Institute trade execution material discusses intraday benchmarks such as VWAP/TWAP and emphasizes that market conditions, volatility, liquidity, spreads, and order characteristics affect trading decisions [4]. Boyd and Busseti also frame VWAP as an execution benchmark and study optimal execution relative to that benchmark [5].

The implication for this ARB engine is simple: if the stock is below fair value but still below/rejecting VWAP or the 200 SMA, the market structure has not confirmed reversion. A confirmed ARB+ should usually reclaim VWAP/SMA or show market-structure proof before printing.

2.4 Pine implementation controls

The code uses TradingView Pine Script v6 features for multi-symbol requests, rolling statistics, VWAP/SMA calculations, and bar/session state controls. TradingView documentation explains that barstate.isconfirmed is true on historical bars and on the closing update of a realtime bar, which can be used to avoid triggering logic before the realtime bar closes [6]. TradingView session documentation also explains how session strings and time() can be used to filter bars by session [7].

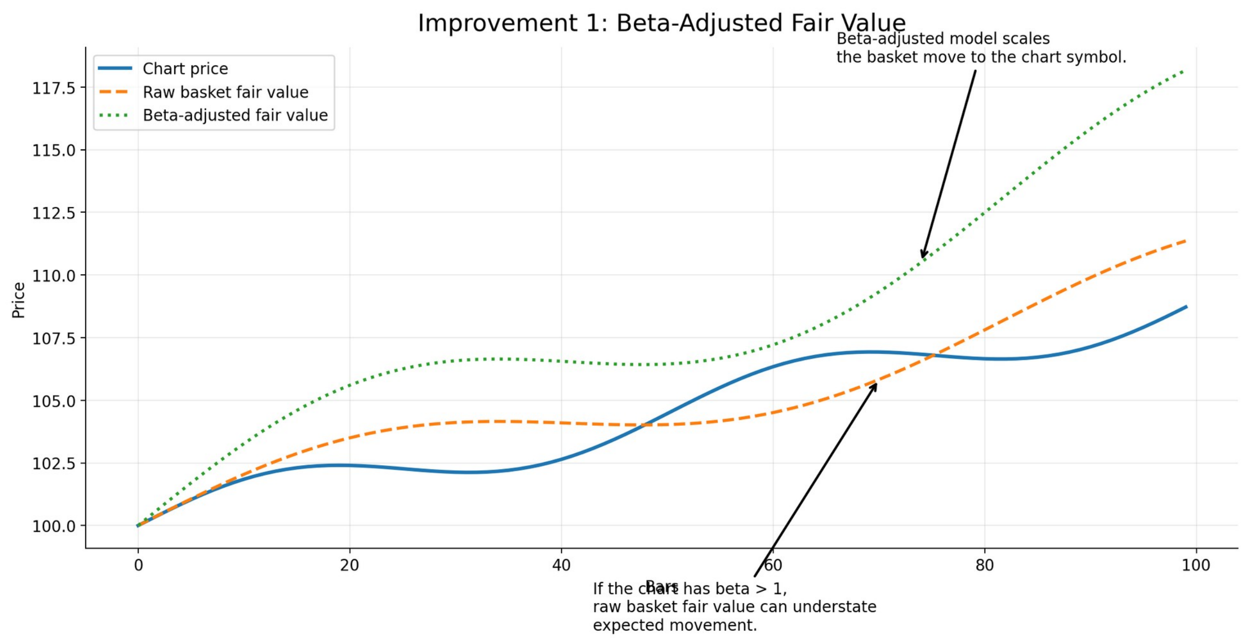

Figure 2. Beta-adjusted fair value scales the basket move to the chart symbol sensitivity.

3. Research Engine v2: What Changed in the Code

The upgraded code is delivered as ARB_Basket_Fair_Value_Research_Engine_v2.pine. It keeps the eight-symbol basket architecture, auto market-cap/manual weighting, synthetic fair value line, z-score model, dislocation markers, dashboard, and alerts. It adds a stricter signal engine designed to reduce false positives near VWAP, SMA, and strong opposing trend structure.

Upgrade	Purpose	Why it helps
Beta-adjusted fair value	Uses rolling beta to scale basket return into expected chart return.	Raw basket return can understate or overstate expected movement when the symbol beta is not near 1.
Beta stability gate	Checks whether beta is stable versus	Avoids using stale or unstable

	its recent average.	sensitivity estimates.
RVOL quality gate	Blocks very low-volume and very high-volume regimes.	Low RVOL can be thin liquidity; high RVOL can be event/forced-flow.
Session filter	Restricts signals to a user-defined regular session.	Reduces extended-hours volume and liquidity distortion.
Bar-close confirmation	Requires the realtime bar to close before the signal becomes true.	Reduces intrabar flicker and live-bar false prints.
VWAP critical gate	Requires reclaim/hold for long or reject/hold for short.	Avoids buying below VWAP while order flow is still bearish.
SMA critical gate	Blocks trades into SMA resistance/support.	Addresses the IBM-style false ARB+ near the 200 SMA.
HTF EMA gate	Optional higher-timeframe trend alignment.	Prevents fading a larger trend without confirmation.
ADX/DMI gate	Avoids strong opposing trend pressure.	Stops ARB+ into strong downtrends or ARB- into strong uptrends.
Expanded confluence score	Aggregates VWAP, SMA, structure, candle, volume, HTF, ADX, and toward-fair evidence.	A signal must earn enough points before printing.

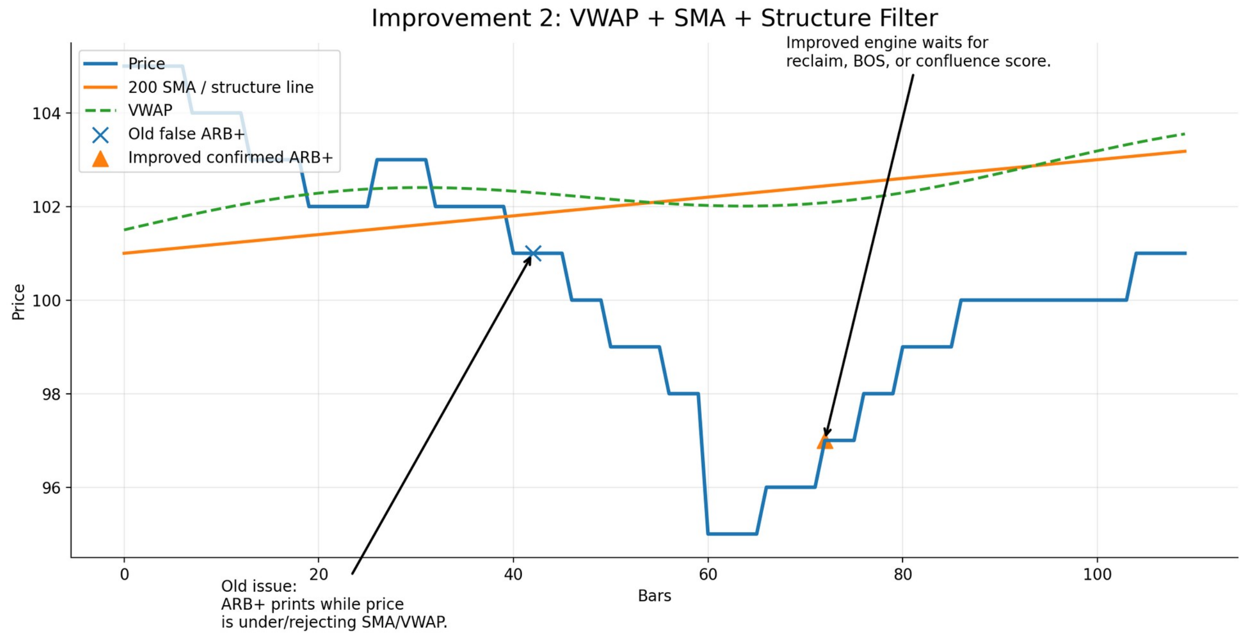

Figure 3. The signal engine waits for VWAP/SMA reclaim or other structural evidence.

4. Definitions and Dashboard Tutorial

Dashboard item	Definition	How to interpret
Dislocation	Whether the residual z-score has crossed the ARB threshold.	YES means watchlist. It does not mean entry.
Basket valid	Correlation, R-squared, beta bounds, active symbols, and beta stability.	FAIL means the basket is not a reliable fair-value anchor.
Beta stable	Checks current beta versus its recent beta average.	FAIL means relationship sensitivity is unstable.
Event/liquidity	RVOL, ATR expansion, gap, z-jump, extreme widening, manual kill, and RVOL quality.	KILL means do not trust mean reversion yet.
Compression	Whether z-score and absolute spread are moving back toward zero.	PASS means cheap is becoming less cheap or rich is becoming less rich.

VWAP	Intraday volume-weighted reference line.	ARB+ wants reclaim/hold above; ARB- wants reject/hold below.
SMA 200	Trend/structure reference line.	ARB+ should not print into unreclaimed SMA resistance.
Structure	Micro break of structure or candle reversal.	PASS means price action confirms the trade direction.
Volume	Minimum/normal RVOL participation check.	Blocks dead liquidity and extreme event volume.
HTF	Higher-timeframe EMA position or slope.	Optional trend alignment filter.
ADX	Trend-stress filter using DMI/ADX.	Blocks ARB+ into strong downtrend or ARB- into strong uptrend.
Session	Whether current bar is in the allowed session.	FAIL means no signal outside the chosen trading window.
Score	Number of enabled confluence items passing.	Signal requires current score >= effective minimum score.
Signal	Final ARB+ / ARB- gate.	PRINT only when all required gates pass.

5. Signal Logic

The final signal logic is intentionally strict. The code does not allow a raw z-score to print ARB+ / ARB- unless the setup passes statistical, regime, compression, and chart-structure gates.

Long-side logic in plain language:

1. The chart must be below basket-implied fair value by enough z-score distance.
2. The basket relationship must be valid: active symbols, correlation, R-squared, beta bounds, and beta stability.
3. The event/liquidity regime must be clear: no gap/ATR/z-shock/manual kill and RVOL must be within quality bounds.
4. The residual must be compressing: z-score and absolute spread move back toward zero.
5. The structure engine must pass: VWAP, SMA, structure, candle, volume, HTF, ADX, and toward-fair evidence reach the score threshold.
6. The signal must satisfy cooldown, session, and optional bar-close confirmation.

Short-side logic is symmetric: the chart must be above fair value, then show evidence that the upside residual is compressing back toward fair value.

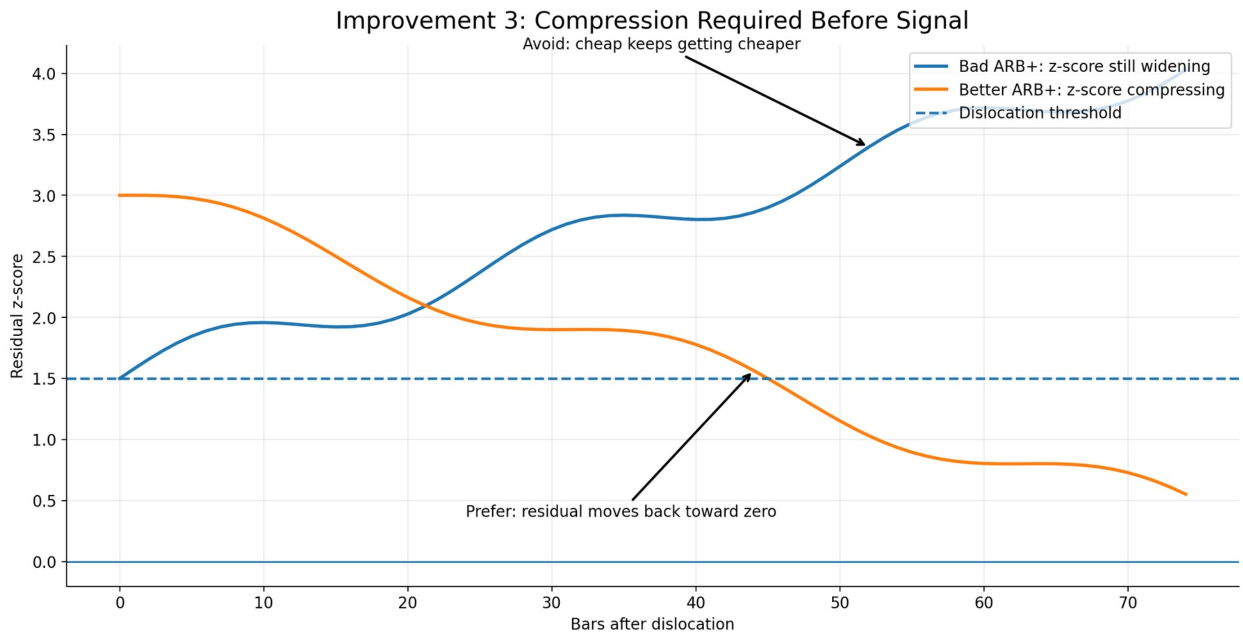

Figure 4. Compression distinguishes a watchlist dislocation from a stronger signal.

Confluence Score: How the Signal Engine Aggregates Evidence

A confirmed ARB is a stacked decision, not a single z-score event.

VWAP	Long: reclaim/hold above. Short: reject/hold below.
SMA 200	Blocks longs under resistance and shorts above support.
Structure	Break of local high/low or reversal candle confirms tape shift.
Volume	Rejects dead liquidity and blow-off event flow.
HTF Trend	Optional higher-timeframe EMA alignment.
ADX/DMI	Avoids fading a strong opposing trend.
Toward Fair	Spread points must shrink toward synthetic fair value.
Candle	Confirmation bar should agree with signal direction.

Default: need 5 of the enabled evidence items, while critical VWAP/SMA/session/bar-close gates must pass.

Figure 5. Confluence score components in Research Engine v2.

6. Practical Tutorial: What To Do When You See ARB+ or ARB-

6.1 When you see D+ or D-

D+ and D- are dislocation-only markers. D+ means the chart is below fair value. D- means the chart is above fair value. Treat these as watchlist alerts. Do not enter just because a D marker appears.

- Check whether Basket valid is PASS. If FAIL, the basket may not explain the stock right now.

- Check Event/liquidity. If KILL, wait until volatility/liquidity normalizes.
- Check Compression. If FAIL, the spread is not yet reverting.
- Check VWAP and SMA. If ARB+ is below/rejecting VWAP or the 200 SMA, wait for reclaim. If ARB- is above/supporting VWAP or SMA, wait for rejection or loss.

6.2 When you see ARB+ blocked

ARB+ blocked means the stock is cheap versus the basket, but one or more gates failed. In the IBM-style failure mode, the 200 SMA or VWAP acts as resistance while price remains weak. The correct action is to wait for reclaim or for the dashboard to change from blocked to confirmed.

Your watchlist conditions for a blocked ARB+ should be: z-score starts falling toward zero, spread points shrink, price stops making lower lows, price reclaims VWAP/SMA, structure flips bullish, and the confluence score reaches the minimum.

6.3 When you see confirmed ARB+

Confirmed ARB+ means the stock is below fair value and the confluence engine agrees that reversion has started. A simple trade plan is: enter on a pullback that holds VWAP/SMA or on the confirming structure candle, stop below the failed reclaim or recent swing low, target halfway to fair value first, then fair value or z-score near zero.

6.4 When you see confirmed ARB-

Confirmed ARB- means the stock is above fair value and the confluence engine agrees that upside excess is compressing. A simple trade plan is: enter after rejection/structure break, stop above the failed breakout or recent swing high, target halfway to fair value first, then fair value or z-score near zero.

7. Settings Guide

Use case	Suggested settings	Trade-off
Balanced intraday	zThresh 1.5-2.0, min score 5, VWAP Reclaim/Reject, SMA Guard + Reclaim, ADX Avoid Strong Opposing Trend.	Good default for 5m-15m charts.
Conservative / fewer false positives	zThresh 2.0-2.5, min score 6-7, require VWAP/SMA, turn on HTF EMA Position, confirm on bar close.	Fewer but cleaner signals.
Fast scalping	zThresh 1.2-1.5, min score 4-5, VWAP Position, SMA Guard Only, structure Micro BOS.	Earlier entries but more noise.
Event/squeeze names	Manual event kill ON during obvious events, RVOL max 4-5, ATR expansion kill 1.5-2.0, require SMA/VWAP.	Protects against CAR-style dislocations.
Extended hours	Either disable session filter intentionally or use a dedicated extended-hours session. Raise minimum RVOL if thin liquidity causes false prints.	Requires separate calibration.

8. Backtesting and Validation Protocol

Do not validate the algorithm only by looking at final ARB labels. Track the full funnel: dislocation -> blocked -> candidate -> confirmed -> outcome. This reveals which gates are doing useful work.

7. Create separate backtest labels for D+ / D-, BLOCK, and confirmed ARB+ / ARB-.
8. Measure how often raw dislocations revert versus how often confirmed signals revert.
9. Record MAE, MFE, time to fair value, time to z-score zero, win rate, R-multiple, and average hold time.
10. Stratify results by Basket valid, Event/liquidity, VWAP, SMA, HTF, ADX, and Score.
11. Run out-of-sample forward testing after tuning. Avoid repeatedly optimizing on the same IBM/CAR examples.
12. Include transaction costs, slippage, spread, and whether the trade happens in regular hours or extended hours.

9. Limitations

- The indicator is not a formal cointegration test. It uses practical rolling proxies that are suitable for Pine but not equivalent to a full econometric model.
- Auto market-cap weights may create a basket that is too mega-cap/AI/growth heavy for some symbols. Manual peer baskets are often better.
- VWAP and SMA filters reduce false positives but can also delay entries or miss V-shaped reversals.
- ADX/HTF filters are optional because some mean-reversion trades intentionally go against trend; use them based on your trading style.
- No indicator can identify all news, borrow stress, halts, dealer positioning, or fundamental shocks. The manual event kill switch remains important.

10. References

[1] Gatev, E., Goetzmann, W. N., and Rouwenhorst, K. G. "Pairs Trading: Performance of a Relative Value Arbitrage Rule." NBER Working Paper 7032, 1999; published in Review of Financial Studies, 2006. https://www.nber.org/papers/w7032

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[3] Krauss, C. "Statistical Arbitrage Pairs Trading Strategies: Review and Outlook." EconStor, 2015. https://www.econstor.eu/handle/10419/116783

[4] CFA Institute. "Trade Strategy and Execution." 2026 refresher reading. https://www.cfainstitute.org/insights/professional-learning/refresher-readings/2026/trade-strategy-execution

[5] Busseti, E., and Boyd, S. "Volume Weighted Average Price Optimal Execution." Stanford, 2015. https://web.stanford.edu/~boyd/papers/vwap_opt_exec.html

[6] TradingView Pine Script documentation. "Bar states." https://www.tradingview.com/pine-script-docs/concepts/bar-states/

[7] TradingView Pine Script documentation. "Sessions."
https://www.tradingview.com/pine-script-docs/v5/concepts/sessions/

Appendix A: Quick Interpretation Card

If you see...	Meaning	Action
D+	Below fair value only.	Watch. Wait for confirmation.
D-	Above fair value only.	Watch. Wait for confirmation.
ARB+ blocked	Cheap but at least one gate failed.	Read the failing row; do not enter by default.
ARB- blocked	Rich but at least one gate failed.	Read the failing row; do not enter by default.
Confirmed ARB+	Long-side relative-value reversion has confluence.	Plan entry, stop, and fair-value targets.
Confirmed ARB-	Short-side relative-value reversion has confluence.	Plan entry, stop, and fair-value targets.

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